

DSA 210 TERM PROJECT

Exploring the Relationship Between Air Quality, Water Pollution, and Economic Investment

1) INTRODUCTION

Environmental pollution is one of the biggest problems the world is facing today. It has a big impact on people's health, nature, and the ability to live sustainably for the future. Even though air and water pollution are often looked at on their own, they are actually linked in many ways. For example, when factories grow or cities expand, it can affect both the air and the water. Some countries that are developing fast and have a lot of money may face environmental issues if they don't have strong rules and healthy ways of managing resources.

The reason for this project is to see if there's a connection between air quality, water pollution, and economic investment. Instead of just looking at one country or using personal data, this project takes a look at many countries around the world. That way, it can compare things more broadly and find ideas that can apply to many places.

The main goal of this project is to check if countries with worse air quality also have more polluted water. It also wants to see how much economic investment, measured as a share of the total economy, is connected to these environmental problems. By mixing environmental and economic data, the project uses data science tools to find trends that might not be clear at first glance.

The following ideas will guide the analysis:

Null Hypothesis (H_0): There is no significant relationship between air quality, water pollution, and economic investment.

Alternative Hypothesis (H_1): Air quality, water pollution, and economic investment are significantly related, and economic factors may help explain variations in environmental pollution.

2) DATA COLLECTION AND DATA SOURCES

This project uses data that is available to the public and is collected from several trustworthy sources.

2.1) Environmental Data Information about air quality and water pollution came from a global dataset that was organized by country.

The dataset includes:

- AirQuality: A score that shows how polluted the air is overall.
- WaterPollution: A number that shows how polluted the water is.
- Country: The name of the country, which helps in organizing and combining the data.

The original environmental dataset had many entries for each country, so we combined the data at the country level. We used the median value instead of the average for both air quality and water pollution. We chose the median because it helps reduce the effect of very high or very low values that might skew the results.

2.2) Economic Data

Economic data came from the World Bank, specifically:

Gross Domestic Investment (GDI), as a percentage of GDP (2021)

This shows what part of a country's economy is spent on investment and is used as a way to measure how active and developed the economy is.

2.3)Data Enrichment

To make the dataset more complete, the environmental data and the World Bank economic data were combined using country names as a link.

Because different sources used slightly different ways to name countries, we made sure all country names were in the same format—lowercase with no extra spaces—before combining them.

3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was done to learn more about how the different variables are spread out and how they relate to each other before creating any formal models.

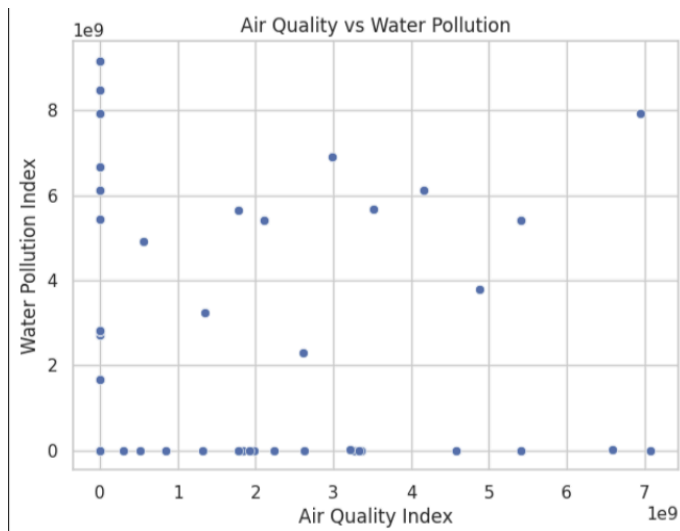
3.1 Distribution of Environmental Indicators

When we looked at the basic facts about the data, we saw big differences in both air quality and water pollution across different countries.

This shows that pollution is affected by many different factors, including the way countries are structured and the policies they follow, not just one single reason.

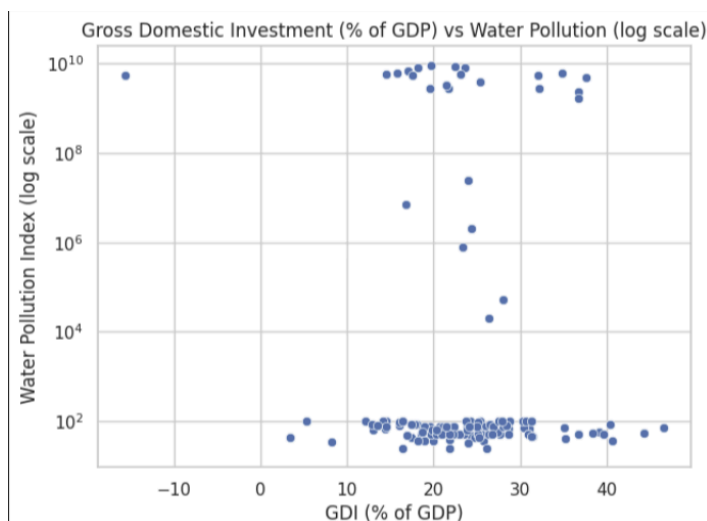
3.2 Air Quality and Water Pollution

To check how air quality and water pollution are connected, we made a scatter plot. This graph shows a somewhat positive link between poor air quality and high water pollution. Countries with worse air quality usually also have more polluted water, but the connection isn't perfect. The way the points are spread out suggests that other things are also important in determining pollution levels.



3.3 Economic Investment and Water Pollution

We also looked at how economic investment relates to water pollution using another scatter plot.



The graph shows a pattern where higher investment doesn't always mean more or less water pollution. Some countries with high investment have low pollution, while others don't. This

suggests that what kind of investment is happening and how strong the environmental rules are are probably important factors.

4. Correlation Analysis

4.1 Pearson Correlation Test (Hypothesis Testing)

To quantitatively assess the relationships observed during the exploratory data analysis, a **Pearson correlation test** was conducted between air quality and water pollution.

- **Null hypothesis (H_0):** There is no linear relationship between air quality and water pollution.
- **Alternative hypothesis (H_1):** There is a linear relationship between air quality and water pollution.

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Pearson correlation coefficient (r): 0.3051598036550353  
p-value: 0.00030331114549509634
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The results show a Pearson correlation coefficient of $r = 0.305$, and the p-value is 0.0003. Because the p-value is less than 0.05, we reject the null hypothesis. This means there is a statistically significant but weak positive connection between air quality and water pollution. Even though this relationship is meaningful, the low correlation coefficient suggests that air quality alone doesn't strongly affect water pollution. This matches the scattered patterns seen in the early plots and shows that other factors, like environmental rules, industry types, and long-term policies, also play a role.

4.2 Correlation Matrix of Environmental and Economic Variables

The correlation matrix reveals the following:

- AirQuality and WaterPollution have a moderate positive correlation (~ 0.31).
- GDI_percent_GDP and WaterPollution show a weak negative correlation (~ -0.13).
- GDI_percent_GDP and AirQuality have a very weak correlation.

These results suggest that air and water pollution are linked, while economic investment alone does not strongly explain pollution outcomes.

6. Machine Learning Analysis

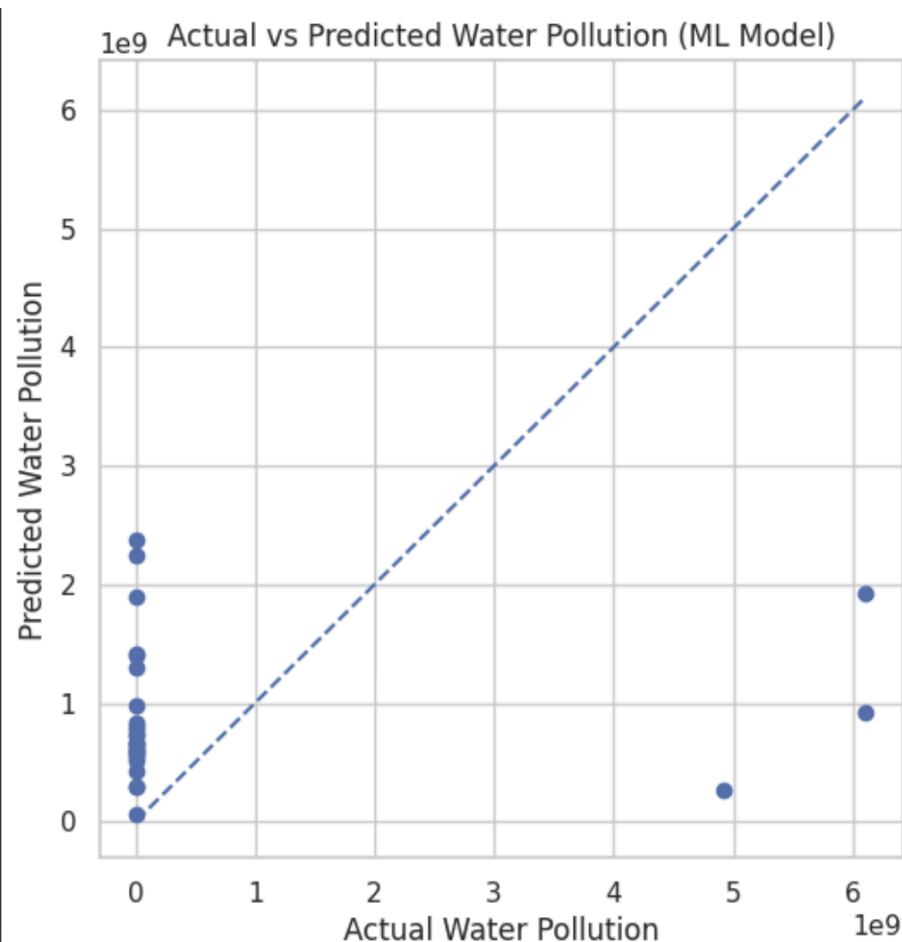
To go beyond the regression analysis, we used a machine learning approach with Linear Regression from scikit-learn.

6.1 Model Setup

- Features: AirQuality, GDI_percent_GDP
- Target: WaterPollution
- Train-test split: 80% of the data was used for training, and 20% for testing

6.2 Model Evaluation

It checked how well the model worked using Mean Squared Error (MSE) and R^2 score. While the model can show general trends, it doesn't do very well with extreme values.



This figure compares real water pollution levels with what the model predicted.

When the points are close to the diagonal line, the predictions are accurate. If they are far away, it shows where the model has trouble. The spread of the points shows that simple models like linear regression cannot fully explain the complexities of environmental pollution. This limitation suggests that non-linear models or additional socio-economic variables could improve predictive performance.

7. Discussion

The results from Exploratory Data Analysis (EDA), correlation analysis, regression, and machine learning give us some important points:

- There is a meaningful relationship between air quality and water pollution at the country level.
- Economic investment does not have a strong direct link with pollution outcomes.
- Simple linear and machine learning models are helpful for spotting trends but not good enough for making exact predictions.

These findings suggest that the quality of policies, how well they are enforced, and how investment is used are probably important factors.

8. Conclusion

This project looked at the connection between environmental pollution and economic investment using detailed country-level data.

By including air quality, water pollution, and World Bank investment data, the study showed that environmental factors are connected, while economic investment has a more complicated role. Even though the models used are simple, they give useful insights and set the stage for future research that could include more social, economic, and policy factors.

AI Usage Statement

Artificial intelligence tools were used during this project to assist with code debugging, data preprocessing guidance, and improving the clarity of written explanations. All data analysis, modeling decisions, and interpretations were designed and executed by the author. AI tools

were used strictly as a supportive resource and did not generate final analytical results independently.